

**Microprocessor Interfacing
and Programming (EE3002)**

Date: December 16th 2024

Course Instructor(s)

Mr. Maaz Rizvi

Final Exam

Total Time (Hrs): 3

Total Marks: 50

Total Questions: 5

Roll No

Section

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Instructions

- Attempt all questions
- Answer all questions on the separate answer sheet
- There are 2 pages of the question paper. Please turn over to see the rest of the questions.
- Make sure your code is neatly written. Excessive cutting/overwriting will be penalized.

Question No. 1 (CLO No. 3)

Marks: 3+7

(A) Write three key differences between a microprocessor and a microcontroller. Which one is more suitable for Embedded applications and why?

(B) Draw the memory map to interface 64 KB of RAM and 256 KB of EPROM with 8086 processor. Also include the number of data lines, and address lines for each RAM and EPROM.

Question No. 2 (CLO No. 4)

Marks: 10

Compose a code in PIC Assembly Language for the following task

The CPU continuously performs the task of transferring data from PORTC to PORTD. After the switch SW1 is pressed, the buzzer will start beeping with a frequency of 0.2 kHz. The buzzer will only go off after completing 300 beep cycles or when SW2 is pressed. (You must show your calculations for the timers)

Question No. 3 (CLO No. 4)

Marks: 3+5

(A) Formulate the pros and cons of using C and Assembly Language to program microcontrollers.

(B) Compose a program in C18 that checks the combination of RB0, RB1, and RB2 and performs the operation on A and B according to the following table

	RB2	RB1	RB0	OPERATION
0	0	0	0	ADD A AND B
1	0	0	1	SUBTRACT A AND B
2	0	1	0	MULTIPLY A AND B
3	0	1	1	DIVIDE A by B
4	1	0	0	INVERT A
5	1	0	1	XOR A AND B
6	1	1	0	AND A AND B
7	1	1	1	OR A AND B

0000 0111 $\Rightarrow$ 7

Question No. 4 (CLO No. 4)

Construct a code in PIC Assembly Language for the following problem statement: The CPU constantly from PORTC to PORTD. A switch is connected to External Interrupt pin to toggle LED 1. Timer0 and Timer 1 toggle LED 2 and LED3 after 10us and 20us respectively. The priority order is as follows

Timer1 >> External Interrupt >> Timer0

Marks: 4+

Question No. 5 (CLO No. 4)

(A) Construct a code in PIC Assembly Language to toggle PORTD 700 times.

7x100

(B) Create a traffic light system in which RED is on for 5m sec, Yellow is on for 2m sec and Green is on for 5m sec. There is a pedestrian button which when pressed will make the red light freeze for 10m sec and then the normal sequence continues for the traffic light.

RA1 → LED3.

RA0 → LED2.

RA3 → LED3.

20ms = (FFFF - 44XX + 1) 0.4μs 10MHz

10ms = (FFFF - 44XX + 1) 0.4μs

25 = (FFFF - 44XX + 1)

44XX = (65536 - 25)

65511 FF E7

0101	0101
0101	1010
0000	1010

20ms = (65536 - 44XX) 0.4μs

50 = 65486

FF CE